



## A Review on Minimum Quantity Lubrication (MQL) Using Hybrid Nanofluids: Enhancing Tool Life and Surface Quality in Metal Cutting

Komal Morankar<sup>1</sup>, Ravindra Deshmukh<sup>2</sup>

<sup>1</sup>Research Scholar, Department of Mechanical Engineering, Jawaharlal Nehru Engineering College, Chh. Sambhajinagar (MS), India.

<sup>2</sup>Professor, Department of Mechanical Engineering, Jawaharlal Nehru Engineering College, Chh. Sambhajinagar (MS), India.

**Abstract:** This review aims to explore the application of vegetable oil based hybrid nanofluids in MQL when machining a high strength steel called EN 24 alloy steel which finds applicability in the manufacturing industry. However, conventional cutting fluids, while cooling and lubricating, create environmental issues making way for MQL and eco-friendly biodegradable oils. By dispersing nanoparticles in a vegetable oil base, hybrid nanofluids exhibit remarkable increase from thermal conductivity to wide-ranging cutting lubrication in machining. The purpose of this article review is twofold: focusing on the major machinability characteristics which are tool wear, surface finish and cutting forces along with evaluating the sustainability influence of hybrid nanofluids at MQL. The outcomes of this research illustrate that hybrid nanofluids decrease the tool wear more significantly than MQL fluids while improving surface finish and reducing cutting force. These enhancements result from formation of a stable tribo-film and a better heat dissipation ability of the hybrid nanofluids and endorse increased tool longevity and improved machining accuracy. Furthermore, the choice of utilizing vegetable oil as base fluid is environmentally friendly since it eradicates generation of toxic waste and pollutes the environment. Based from the discussions made in this review, there are some areas that needs to fill up by new research as follow: economic analysis and long-term performance evaluation. In the course of this research, there are the following recommendations on how future research could provide the best practicable use of hybrid nanofluids in sustainable machining.

**Keywords:** Minimum Quantity Lubrication (MQL), Hybrid Nanofluids, Machinability, Tool Wear, EN 24 Alloy Steel.

### 1. Introduction

Selection of machining methods and cutting fluids has emerged as one of the most important research themes in the era of modern manufacturing towards sustainability. Machining typical machining processes involve large amounts of synthetic cutting fluids to minimize friction and eliminate heat, generated from cutting operations especially in tough materials such as the EN 24 alloy Steel [1]. However, the applications of these fluids are inline with following environmental issues, such as toxicity of the fluid, a problem of its disposal and the fact that the basic fluid is derived from a non-renewable resource [2]. As such, industries and researchers have been forced to look for environmentally friendly counters as a way of minimizing the use of lubricants and any adverse effects that they may have on the environment through the use of a technique such as Minimum Quantity Lubrication[3].

Minimum Quantity Lubrication or MQL is emerging as a viable and environmentally friendly approach to flood cooling. MQL is a technology whereby a minimal amount of cutting fluid is sprayed at the cutting

area in an attempt to use considerably less of the fluid but ensure the area requires is adequately lubes and cooled [4]. Nonetheless, MQL by itself may not be sufficient to provide the cooling requirement for high speed and high strength cutting operation, thus researchers are looking into nanofluids which are fluids containing nanoparticle to improve the ability of MQL. Several researchers involve the use of nanoparticles, particularly  $\text{Al}_2\text{O}_3$ ,  $\text{TiO}_2$ , and graphene, in cutting fluids and their results indicate that enhancing heat transfer and lubricating properties considerably reduces tool wear, surface roughness and cutting forces [5].

But the possibilities of further improvement of nanofluid characteristics do not end here: materials aliased as hybrid nanofluid are mixtures of different nanoparticles in the base fluid [6]. These fluids utilise a combination of multiple nanoparticles to realise improved thermal conductivity, minimized friction, and actualisation of a stable tribo-film on the tool-work piece interface thereby minimising wear on the tool. In addition, it is proposed to use vegetable oil as a base fluid, which already includes biodegradability and renewability of the cutting process in the context of sustainable manufacturing objectives [7].

In this review, only vegetable oil based hybrid nanofluids for MQL turning process of EN 24 alloy steel are highlighted since it is used broadly in automotive, aerospace and heavy industries for its high tensile strength and toughness. EN 24 steel is hard to machine for the fact that it is highly hard and then leads to high rate tool wear thus demanding efficient lubrication and cooling [8]. The current literature review in this paper will focus on recent development in hybrid nanofluids and their influence on the machinability of the materials with respect to tool wear, surface roughness, and cutting forces [9]. Furthermore, it covers the advantages and disadvantage of sustainability of the said fluids, incorporated the gap and potential research to capture and enhance the reality of these fluids in industry [10]. The aims and scope of the present work are as follows: To discuss and examine the recent development of hybrid nanofluids for sustainable machining, To evaluate the effectiveness of MQL with nanofluids, and To identify research opportunities for improvement of the performance of hybrid nanofluids for challenging materials entered in present days like EN 24 alloy steel [11].

### **1.1 MQL and MQCL Technology**

Significantly, researchers have become familiar with a new concept called Minimum Quantity Lubrication (MQL) owing to issues with biological safety, as well as the environmental deterioration in connection with machining processes. Known also as “Near-dry machining” or “Micro lubrication,” MQL is also aimed at the least possible usage of cutting fluids, which brings obvious economic advantages as the costs of the liquid for forming a lubrication film are reduced [13]. MQL delivers a small, accurately measurable amount of cutting fluid at the contact point in between the tool and the work piece [14]. However, while MQL reduces fluid consumption it does has some drawbacks a example limited fluid approach to the machining zone might be affected. Another system used in machining operations is the mist lubrication which has been proved to have better results using both lubricant and air

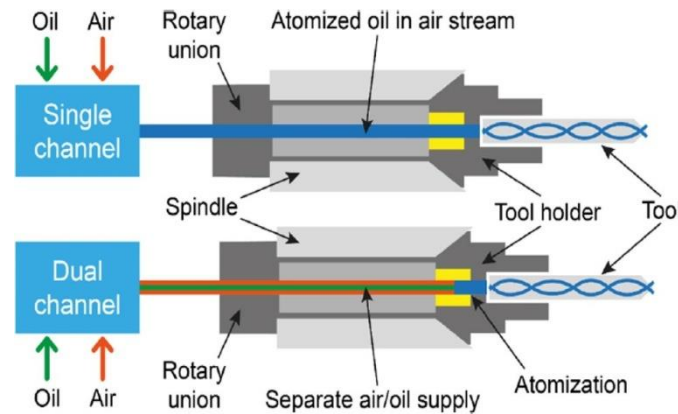
in the machining zone at an increased flow rate and pressure [15]. The method of using water based microjet impingement or MQL technique is superior to common wet cooling methods due to its improved cooling through heat transfer by evaporative and convective methods. Some works reported that high-velocity fluid droplets can erode the vapor barriers at the tool workpiece interface, thus increasing heat transfer and minimizing mechanical energy dissipation. Through the use of MQL, the process effects adequate amounts of oil droplets for lubrication, and compressed air for chip removal hence makes the procedure to be effective in minimizing friction thus enhancing the tool's life [16]. However, the MQL method cannot eliminate some disadvantages because of the poor penetrability of the aerosols, hence limiting the boundary lubrication at the tool work piece interface besides, small droplets evaporize and offer small coolant effect. However, MQL technology incorporates the Rebinder effect that decrease the hardness and ductility of the workpiece and develop a molecular film on the finished surface. The impact accomplishes a similar goal in the material by promoting elevated plastic flow for the sake of better machinability. Some findings from related studies suggest that oil particles stick to the machined surface to allow easy plastic deformation of the chips enhancing overall machinability [17].

MQL parameters such as location of nozzles also affect the machining performance in a considerable way. For instance, the direction of the nozzle affects the extent to which the cutting fluids get into the machining area. Other researches point out that when the nozzle is placed at  $45^\circ$  with the horizontal and the vertical planes, it minimizes friction best [18]. Another important parameter is pressure of air, which affects tool's life and quality of the surface. It was found out from experiments that maximum pressure of air reduced tool life up to the same level as dry machining, even though the surface finish may be better at higher pressures [19]. The quantity of cutting fluid also influences the tool life where increased flow rate of cutting fluid improve the tool longevity although without considerable improvement to the surface finish. Minimum Quantity Cooling Lubrication (MQCL), an extended version of MQL, uses both, cooling and lubrication. In MQCL, an emulsion is usually employed in place of oil in order to cool and lubricate the system at the same time. This method has proved to provide positive outcomes towards tool wear protection and proper chip formation. Further, cold mist cooling has been applied in some applications like in machining of titanium alloys the cooling is optimized using cold mist cooling systems. MQL and MQCL can be communicated with either internal as well as external manned system. In internal systems, air and lubricant are injected into the nozzle and delivered toward the machining zone using single or dual channels most of the time. The externals systems require an atomizer to help it to prepare and place the aerosol over the machining region. There are two main types of nozzle: the ejector nozzle and the conventional nozzle and each type can be chosen according to the needs of a particular process [20].

Recently, advancements in MQL technology have introduced time-controlled MQL, where pulsing intervals of lubricant application are managed to improve machining outcomes, particularly in terms of

surface finish. This evolution represents the current state-of-the-art in MQL technology, enabling precise control over lubricant application, optimizing surface quality, and further enhancing the efficacy of MQL in machining applications [21].

This comprehensive approach aligns with efforts to enhance operational performance and sustainability in manufacturing, echoing the goals of minimizing environmental impact, improving product quality, and reducing machining costs [22].



**Figure:** MQL internal delivery system[23]

### 1.3 Significance of MQL in the Machining Domain

Minimum Quantity Lubrication (MQL) technology in machining has been studied before and it has been shown that the technology enhances the cutting performance when dealing with materials. The present study for MQL has demonstrated improved machining of difficult-to-cut materials including titanium and stainless steel as it provides numerous advantages leading to better results in machining [24]. This gives MQL technology right of top priority owing to the fact that one of the major issues associated with machining is that it puts through high cutting forces in contrast to the wet as well as dry strategies. This study shows that when the flow rate of MQL is optimal, the magnitude of the cutting forces in its three directional dimensions is reduced making the machining process easier [25]. Nevertheless, after a certain flow rate this reduction of the cutting forces prove to be steady, implying that certain flow parameters are ideal for the best results to be obtained. MQL also means priceless advantages in minimising the cutting temperate and which in turn retains the sharpness of the tool and has a better interaction between the cutter and chip [26]. Although it cannot cool like a regular flooded cooling system, MQL is very efficient in cooling the heat generated in the cutting zone than dry machining. From an operational perspective, MQL is also advantageous in cost and demand of energy hence offers a better shot at economical machining. Demonstrating significant improvements in cutting performance across various materials. MQL has been shown to enhance the machining of challenging materials such as titanium alloys and stainless steel, delivering a range of benefits that improve machining outcomes. One of the primary advantages of MQL technology is its ability to significantly reduce cutting forces compared to traditional wet and dry machining methods. Research indicates that with an optimal MQL flow rate, the cutting force

components are minimized, enhancing the ease of machining. However, beyond a specific flow rate, this reduction in cutting forces tends to stabilize, suggesting that optimal flow parameters are necessary to achieve the best performance. MQL also offers substantial benefits in reducing cutting temperature, which helps maintain tool sharpness and improve the interaction between the tool and the chip. While MQL cannot achieve the same cooling effects as conventional flooded cooling systems, it is still highly effective in controlling heat generation at the cutting zone, especially when compared to dry machining. From an operational standpoint, MQL is also superior in terms of cost-effectiveness and energy consumption, making it an economical choice for sustainable machining. Concerning the improvement of surface finish, MQL performs a magnificent role, as surface finish is almost similar to that of flooded lubrication and far better than dry machining. In addition, MQL also causes less wear on tool and increases dimension precision; therefore, enhancing the quality of the final products. Several qualitative parameters which characterize MQL have been identified and include the nozzle distance and the volumetric air flow. Changing these factors determines the size of the lubricant droplets and the wetting area that affect the rate of lubrication and cooling at the tool chip interface area [27].

This cumulative evidence emphasizes the importance of MQL technology in modern machining, aligning well with goals of improving efficiency, reducing operational costs, and minimizing environmental impact. By optimizing key MQL parameters, manufacturers can achieve substantial gains in tool life, surface quality, and overall machining performance, which are critical for meeting stringent quality standards and maintaining sustainable practices in the manufacturing industry [28].

#### **1.4 MQL and Sustainability**

With the advancement of industries the emission of harmful and hazardous wastes is on the rise causing harm to the environment and health of the people, pointing towards the efficient and ecofriendly production. Owing to regulatory requirements and programs that call for a safe manufacturing condition this shift has become imperative to modern industry [29]. Sustainability in manufacturing cannot be realized without methods that seek to conserve energy and implement ways to recycle industrial wastes, utilize resources, and use better lubrication in addition to preserving natural resources. As recommended by safety entities like OSHA, NIOSH, and EPA, researchers and industries are now giving attention to new generation metal cutting fluids to reduce the use of dangerous fluids to the greatest extent possible. Minimum Quantity Lubrication (MQL) has turned out to be a preferred solution to the traditional approaches to lubrication such as flood lubrication that give rise to problems such as generation of mist. MQL had a high degree of impact on cleaner machining environments; it has less consumption of cutting fluids and made the operation more sustainable for manufacturing, thus, the tool was found efficient for sustainable manufacturing. Despite this capability to radically improve Metal Cutting in a sustainable way, the applicability of MQL in the industries remains somewhat restrained. A lot of companies are striving to design sustainable manufacturing strategies, but more cooperation is needed for better link

between the global sustainable development policies and real practical achievements. Therefore to achieve higher sustainability goals, it is crucial that MQL technology gets adopted all over the manufacturing sectors. The engineering alloys are strong, rigid and are difficult to process and hence the study of their machinability has recently attracted lots of attention [30]. A host of experimental research suggest that MQL increases the machinability of these materials and underlines the role of MQL in making metalworking environmentally friendly. Furthermore, the contextual analysis of MQL assisted machining strategies identifies potential fields of application of this technology, confirming the role of MQL in the ability to achieve constant and effective machining in the future [31].

## **2. Literature Review**

**Naresh Babu et al. (2018)** conducted an experimental study on the effectiveness of Minimum Quantity Lubrication (MQL) using copper nanofluids in turning processes. Using H13 steel with an uncoated tungsten carbide tool, they varied cutting speed, environment, and feed rate as input parameters to measure output responses like temperature, surface roughness, and tool wear. The study utilized VIKOR as the decision-making method and found that copper nanofluids in MQL notably reduced surface roughness and tool wear, providing enhanced machinability through better heat dissipation and lubrication.

**Chetan et al. (2016)** explored the use of nanofluids in MQL for turning operations with multilayered WC inserts on Nimonic 90. They employed  $\text{Al}_2\text{O}_3$  nanoparticles in sunflower oil and silver (Ag) nanoparticles in water at varying concentrations (5%, 10%, and 15%). Using Response Surface Methodology (RSM) for optimization, the study analyzed speed, feed, and depth of cut as input parameters and found that  $\text{Al}_2\text{O}_3$  nanoparticles produced the lowest cutting force, surface roughness, and tool wear due to a tribo-film formation. This tribo-film effectively reduced sliding friction, improving the machining performance.

**Devinder Priyadarshi et al. (2015)** investigated the optimization of cutting parameters in the turning of Al-6061-SiC-Gr hybrid nanocomposites using RSM. Focusing on cutting speed and feed rate, they measured cutting force and surface roughness to assess the material's machinability. The study concluded that feed rate had the most significant influence on cutting force, followed by depth of cut, while cutting speed had an insignificant impact on cutting force but positively affected surface finish. This research emphasizes the importance of adjusting feed rate to optimize machinability when working with hybrid nanocomposites.

**Amiril Sahab et al. (2018)** studied the effect of modified vegetable oils on tool failure and wear mechanisms in turning AISI 1045 medium carbon steel. With speed, feed, and depth of cut as input parameters, the study measured tool life, tool wear, and surface roughness using a modified jatropha oil enhanced with ionic liquids. Results indicated that this modified oil (MJO+AIL) substantially reduced

surface roughness, tool rake, flank wear, and notch wear, underscoring the effectiveness of vegetable oil-based lubricants in enhancing tool longevity and workpiece quality.

**Das et al. (2019)** performed a performance evaluation of various cutting fluids in MQL hard turning of AISI 4340 steel using an uncoated cermet insert. They examined the effects of cutting speed, feed rate, and depth of cut on feed force, cutting force, radial force, and tool wear. Statistical analysis showed that nanofluids containing  $\text{Al}_2\text{O}_3$  nanoparticles outperformed both water-soluble coolant and compressed air, significantly reducing tool wear and cutting forces. This research highlights the superior performance of nanofluids in MQL applications, particularly in hard machining scenarios.

**Padmini et al. (2018)** conducted an experimental evaluation of nano-molybdenum disulfide ( $\text{MoS}_2$ ) and nano-boric acid suspensions in vegetable oils as potential cutting fluids for turning AISI 1040 steel. Using MQL, they measured temperature, tool flank wear, and surface roughness with varying speeds, feed, and depth of cut. The study found that the combination of  $\text{MoS}_2$  and boric acid in vegetable oil at specific concentrations was highly effective in reducing cutting forces and temperatures, resulting in lower tool wear. This research underscores the potential of hybrid nanofluids in enhancing both cooling and lubrication during machining.

**Mohsen Khajehzadeh et al. (2019)** examined the influence of nanolubricant particle size on flank wear in the hard turning of AISI 4140 steel. Using a tungsten carbide tool coated with TiCN and  $\text{Al}_2\text{O}_3$  layers, they analyzed the effects of cutting speed, particle concentration, and particle size on cutting tool temperature, cutting force, and tool wear. The study compared MQL, Minimum Quantity Coolant Lubrication (MQCL), Dry Flooded, and Nanofluid Minimum Quantity Lubrication (NMQL) conditions. Results indicated that smaller nanoparticles in NMQL reduced flank wear and cutting force more effectively than larger particles, highlighting the importance of particle size optimization in nanolubricants for improving tool life in hard turning.

**Anuj K. Sharma et al. (2017)** investigated the use of alumina ( $\text{Al}_2\text{O}_3$ ) and molybdenum disulfide ( $\text{MoS}_2$ ) hybrid nanoparticles in vegetable oil-based cutting fluids for MQL hard turning of AISI 304 steel. They varied cutting speed, feed, depth of cut, and nanoparticle volume concentration (0.25%, 0.75%, and 1.25%) to assess their impact on cutting force, thrust force, feed force, and surface roughness. Using Response Surface Methodology (RSM) for analysis, the study reported an 8.4% improvement in machining performance due to the hybridization of  $\text{Al}_2\text{O}_3$  with  $\text{MoS}_2$ . However, the combination negatively affected thermal conductivity, suggesting a trade-off between lubrication benefits and thermal management when using these hybrid nanofluids.

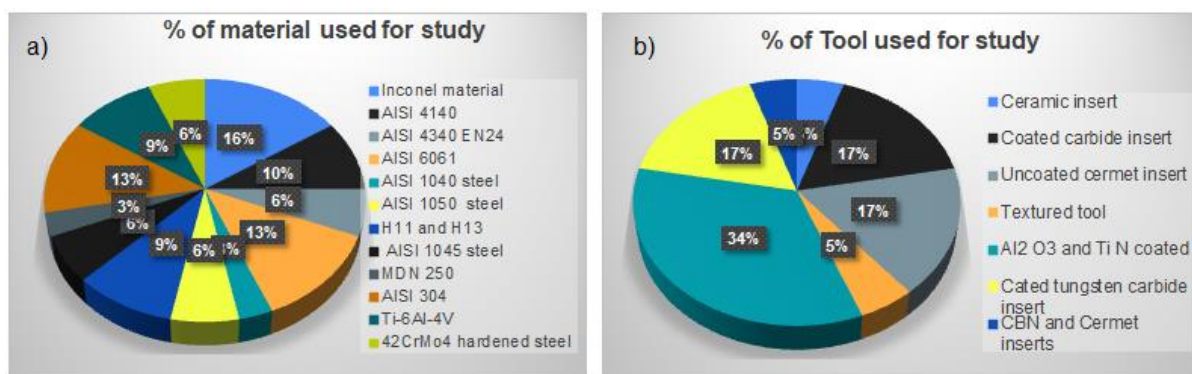
## 2.1 Conclusion of Literature Review

**Summary of Findings:** The two literatures that compare Minimum Quantity Lubrication (MQL) and nanofluids reveal that employing lubricants that incorporate nanoparticles in the process of machining endows the tools with longer life, and also causes the surface roughness and intensities of cutting forces

to be reduced. Research has it that tool surfaces in contact with nanofluids – including those containing multiple nanoparticles such as  $\text{Al}_2\text{O}_3$  and  $\text{MoS}_2$  develop a stable tribo-film to mitigate the forces of friction and wear. Furthermore, vegetable oil based nanofluids have been tested and found to present good potentials in attaining environmentally friendly machining through offering optimum cooling and lubricating properties besides being biodegradable and nontoxic. However, the effectiveness of these fluids is much related to such characteristics as nanoparticle size, nanoparticle concentration, and the type of nanofluid used due to their thermal and lubricant performances.

**Current Trends:** Heterogeneous nanofluids that consider more nanoparticles are studied bringing the perks of both types to the MQL process and the improvement of the thermal conductivity of the lubricant. Hybrid nanofluids research is shifting to vegetable oil as the base fluid due to sustainable development initiatives in manufacturing. The techniques such as Response Surface Methodology (RSM) and statistical analysis are mostly used to enhance the cutting performance of machining parameters, especially in MQL studies. **Limitations:** Despite this, some limitations can still be identified in the current research that can be filled in some way by future studies. To the best of author's knowledge, no comprehensive research work has been reported on vegetable oil based hybrid nanofluid for MQL in machining EN 24 alloys steel. Also, long-term economic analysis and studies comparing the efficiency of conventional nanofluids and hybrid nanofluids are scarce; therefore, there is doubt about hybrid nanofluid practicality for the broad utilization in industrial applications. These limitations point towards the need for a future study directions on the most efficient formulation of hybrid nanofluids, sustainability indices, and cost analysis of the fluids under the different machining conditions.

### 3. Analysis of Materials and Tools Studied in MQL Machining Research



**Figure 2:** MQL Studies a) Percentage of materials used a) Percentage of tool used

Figure 2 illustrate the types of materials and cutting tools commonly studied in the context of Minimum Quantity Lubrication (MQL) and nanofluid-assisted machining. Analyzing this data offers insights into current research trends and highlights areas that could benefit from further exploration [32-35].

**Materials Used in MQL Studies:** shows the diversity of materials researchers have investigated. A significant portion of studies (16%) focus on Inconel alloys, which are valued for their high temperature and corrosion resistance but are notoriously difficult to machine. The next most commonly studied



materials are AISI steels—notably AISI 4340 (10%) and AISI 4140 (9%) which are widely used for high-strength applications in manufacturing. Other AISI steels, such as AISI 1040 (9%) and AISI 1045 (6%), offer a balance of hardness and machinability, making them ideal for MQL studies. Additionally, AISI 304 (9%) stainless steel is often chosen due to its corrosion resistance but relatively poor machinability, which makes it an excellent candidate for evaluating MQL's performance benefits.

The chart also highlights less frequently studied materials, such as AISI 6061 aluminum (6%) and MDN 250 steel (13%). These represent areas where further research could yield valuable insights, especially regarding the benefits of MQL in lighter and high-strength precipitation-hardening materials.

**Tools Used in MQL Studies:** provides a breakdown of the types of cutting tools used in MQL machining research. The majority of studies (34%) use coated carbide inserts, which are highly effective in high-temperature machining due to their wear resistance and thermal stability. The coatings, typically TiN and  $\text{Al}_2\text{O}_3$ , enhance durability and reduce friction, making these tools well-suited for MQL conditions. Other popular choices include CBN and cermet inserts (17%), as well as  $\text{Al}_2\text{O}_3$ -coated inserts (17%), both known for their hardness and temperature resistance in high-speed machining.

Textured tools (5%) and uncoated cermet inserts (5%) appear less frequently in studies. Textured tools are of particular interest as they improve lubricant retention, which could enhance MQL effectiveness. The use of ceramic inserts (17%) also demonstrates an interest in materials that can withstand high temperatures, ideal for hard turning applications.

#### Summary of Findings and Research Gaps

The data shows a strong focus on high-strength, heat-resistant materials and coated tools in MQL research. While these studies advance knowledge in machining challenging materials, there are still gaps. For example, materials like aluminum and uncoated tools are underrepresented, indicating potential areas for further investigation. This analysis highlights opportunities to expand MQL research to encompass a broader range of materials and tool types, particularly in exploring sustainable machining solutions with less conventional materials and uncoated tools.

## 4. Discussion: Implications for Machinability and Sustainability

### 4.1 Benefits of Hybrid Nanofluids

Hybrid nanofluids have been applied in machining with much positive impacts such as reduced tool wear, enhanced surface quality, and low cutting force. These nanofluids are used to form strong and stable tribo-films on the tool surface in order to reduce fulfill the requirement of low coefficient of friction as well as providing a shield on the cutting edges where the tool is used hence increasing its life span. This improvement is especially desirable in critical applications like machining EN 24 alloy steel where higher strength and wear properties are beneficial. Furthermore, in terms of the base fluid in these hybrid nanofluids it is worth mentioning that vegetable oils are less hazardous as compared to the synthetic ones that are usually used in engineering. This is not only good for business since it allows for the

enhancement of sustainability but also because of the increasing regulation of environment harm limitation.

## **4.2 Environmental and Economic Considerations**

Although the hybrid nanofluids provide a range of machinery advantages the problems related to the machining cost and large scale application persists. However, the higher first costs of hybrid nanofluids could dissuade application at first instance, especially for small and medium-scale manufacturing industries. To support the change, however, it is necessary to calculate long term economic impacts all necessary to prove their efficiency in the long term. Further, an extensive assessment of the environmental effects and cost-benefit analysis is required to determine if the hybrid nanofluid offers more benefits to embrace sustainable development than the initial cost to establish it.

## **5. Conclusion and Future work**

The present study mainly relates to the application of Vegetable oil-based hybrid nanofluids in Minimum Quantity Lubrication (MQL) systems, featuring improvements in machining high strength materials like EN 24 alloy steel. These hybrid nanofluids improve the machinability aspects of a workpiece by increasing tool durability, better surface quality, and reduction in cutting forces. Apart from being an environment friendly solution of traditional cutting fluids this option also helps to meet environmental standards of modern manufacturing processes. However, to realise the maximum potential of hybrid nanofluids in industrial applications, there requires further studies. Research and development of the economic profitability for the use of the technology and its effect on the environment as well as the most suitable fluid composition must be conducted to make the consumption of the technology economic. , Moreover, more careful studies regarding the interactive effects of the material, tool and Nano fluid will be of paramount importance to set the ground for this advantageous technology in sustainable machining practices.

### **5.1 Future Research Directions**

This review highlights several critical areas for advancing hybrid nanofluid research: Long-Term Economic Feasibility: There is still a lack of detailed cost benefit assessment of the application of hybrid nanofluids on a larger scale in the industrial sector. To these studies, both the capital cost of incorporating these technologies as well as the total life-cycle cost of using the technologies should be included in order to properly factor in the gains that possibly come from longer tool life and easier machinability. Optimized Nanoparticle Formulations: Further research should be aimed at understanding which nanoparticle parameters like size, concentration and composition should be used to achieve the right balance of both lubricating and thermal applications of nanoparticles. This optimization will also enhance the performance of the fluid and expansion of its usability of various machining processes. Tool Compatibility Studies: Correct comprehension of the mechanics by which these hybrid nanoflids perform with diverse tool kinds and coatings, namely uncoated or textured tools, is critical for MQL optimality.

Studying these interactions will contribute to further development of the MQL strategies for all kinds of tooling applications.

**Specific Applications in EN 24 Alloy Steel:** Given the scarcity of targeted studies on EN 24 alloy steel, it is important to investigate the specific effects of hybrid nanofluids on machinability parameters for this material. Studies that focus on EN 24 alloy can provide valuable insights into the fluid's impact on critical outcomes like tool wear, surface quality, and cutting forces, further supporting its adoption in challenging machining tasks.

#### **Authors' Contribution:**

**Komal S. Morankar** Conducted the literature review, previous research paper studied data analysis, and manuscript writing and **Dr. Ravindra R. Deshmukh** supervised reviewed and finally approved the manuscript.

#### **Conflicts of Interest:**

The authors declare no conflicts of interest about the publication of this review paper.

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